

LAGRANGE: A Proposal for Literature-Adaptive Generation and Assessment of Scientific Research Questions

Abstract

This proposal describes an end-to-end system for generating and assessing scientific research questions, with an initial focus on physics-adjacent domains. The proposed pipeline couples literature-grounded question generation with adaptive question assessment. On the generation side, the system uses `deep-literature-search` to assemble a topic-specific literature report and evidence graph, then uses `question-formulator-beta` to synthesize and refine candidate research questions. On the assessment side, it extends the ArgumentBankGen paradigm (see Algorithm) similar to Zhou et al. [2024] with literature conditioning inspired by Liu et al. [2025], review-derived weak supervision from OpenReview, and dynamic inference policies that go beyond a static hypothesis bank. The key idea is that research-question quality is partly global and partly local: some criteria, such as specificity or falsifiability, are stable across domains, while others, especially novelty and feasibility, depend on the current state of the literature frontier. Accordingly, the proposed assessor combines three sources of evidence: reviewer rationales extracted from historical reviews, literature-grounded arguments derived from current retrieval, and query-conditioned arguments synthesized at inference time. Rather than only predicting a binary *good/bad* label, the system produces a multidimensional quality profile, an overall ranking score, and a short critique that can be used to revise the question before final output. The proposal argues that this hybrid design should yield a more reliable, interpretable, and scientifically useful research-question generator than either one-shot prompting or static-bank inference alone.

1 Introduction

Formulating a good scientific research question is an unusually difficult cognitive task. A useful question must be novel relative to prior work, concrete enough to guide computation or experiment, narrow enough to be answerable, and important enough to justify follow-up. These requirements are hard to satisfy simultaneously, which is one reason research ideation remains a bottleneck even in domains where large language models (LLMs) are already helpful for summarization and drafting. Recent work has shown that LLMs can generate hypotheses from either data or literature, but question quality remains difficult to assess in a principled way [Zhou et al., 2024, Liu et al., 2025, Vasu et al., 2025]. In practice, most current systems still rely on one-shot prompting, proxy task accuracy, or loosely defined reviewer-style judgments.

The internal generation stack already provides a natural starting point for a more rigorous system. `deep-literature-search` can produce a topic-conditioned literature report and evidence structure, while `question-formulator-beta` can use that report to formulate and refine a technically specific question. What is missing is an assessor that can decide whether the resulting question is actually *good* in a scientific sense. This assessment problem is not merely stylistic. A question can read fluently while being trivial, already answered, too broad to test, infeasible under current methods, or

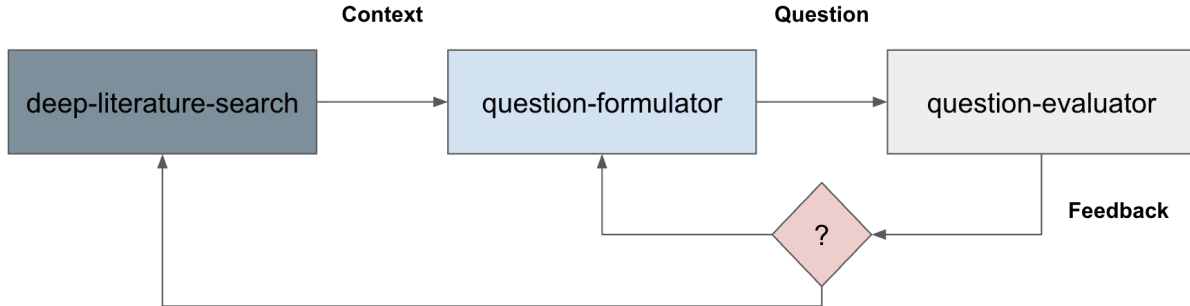


Figure 1: High-level overview of the proposed research-question generation and assessment pipeline. **deep-literature-search** produces topic-specific context, which is passed to **question-formulator** to generate a candidate research question. The **question-evaluator** then assesses the question and returns feedback. This feedback can be used either to revise the question through the formulator or to trigger an updated literature search, yielding a closed-loop system for generation, evaluation, and refinement.

disconnected from the literature frontier. Conversely, a high-quality question often has a distinctive structure: it states a mechanism, specifies variables and observables, defines the regime of interest, and makes falsification possible. The provided example question illustrates precisely this target form, namely a parameterized, mechanistic, and testable question rather than a generic research theme.

A naive solution would be to cast assessment as binary classification using accepted versus rejected papers. That would be useful, but it would also be insufficient. Accept/reject outcomes are noisy because they conflate underlying question quality with execution quality, writing quality, venue fit, reviewer disagreement, and resource constraints. In addition, a purely static classifier will struggle with the moving nature of novelty: whether a question is interesting depends on what the literature currently says. This is why the central design choice in this proposal is to treat question assessment as *literature-adaptive* and *review-supervised*. Historical reviews provide weak supervision about durable scientific virtues, while current literature retrieval provides the local evidence needed to judge novelty and feasibility for a particular topic.

We therefore propose LAGRANGE, an end-to-end generator–assessor–reviser for scientific research questions. The generator uses **deep-literature-search** and **question-formulator-beta** to produce several candidate questions. The assessor extends HypoGeniC-style argument-bank learning [Zhou et al., 2024] with the literature–data synergy of Liu et al. [2025]. Instead of relying on a single static bank of arguments, it merges (i) reviewer-derived arguments learned from OpenReview data, (ii) literature-derived arguments grounded in the retrieved topic graph, and (iii) a query-conditioned dynamic bank generated at inference time. The resulting system does not simply assign a label; it returns a quality vector over novelty, significance, specificity, grounding, feasibility, and falsifiability, along with critiques that can be used to revise the question. The proposal is motivated by a simple hypothesis: research-question assessment works best when general scientific norms learned from historical review behavior are combined with frontier-specific evidence from the current literature.

Figure 1 illustrates the overall architecture of the proposed system. The pipeline begins with **deep-literature-search**, which constructs literature-grounded context for a target topic. This

context is consumed by `question-formulator` to produce a candidate research question. The candidate is then passed to `question-evaluator`, which assesses its quality and generates feedback. Importantly, the feedback closes the loop: it can be used either to refine the question directly or to request additional literature retrieval, enabling an iterative process in which generation and assessment are jointly improved.

2 Related Work

Recent work on LLM-based hypothesis generation divides naturally into data-driven and literature-driven approaches. Zhou et al. [2024] introduced a data-driven formulation in which an LLM generates and updates a bank of natural-language hypotheses that improve classification performance. Their central insight is that a bank of interpretable hypotheses can be updated online through an exploration–exploitation process, and that adaptive inference over the bank can outperform few-shot prompting. This provides the conceptual starting point for the assessment module proposed here: instead of treating reviewer arguments as static prompts, we can treat them as hypotheses about what makes a research question strong or weak.

The next step in the literature is to combine data-driven and literature-driven evidence. Liu et al. [2025] show that literature-based and data-based hypothesis generation are complementary rather than redundant. Their hybrid method refines data-driven hypotheses using literature summaries and also studies direct bank union after redundancy reduction. This result is highly relevant to research-question assessment because the notion of a good question is inherently hybrid. Some criteria are general and can be learned from historical examples, but novelty and feasibility are only meaningful relative to the current literature. In parallel, Vasu et al. [2025] argue that literature-grounded hypothesis generation should not stop at conditional text generation; it should explicitly validate reasoning chains and preserve provenance. That emphasis on evidence-backed reasoning directly informs the proposal below, especially the requirement that every assessment should cite concrete literature-based support.

At a broader systems level, recent work has explored increasingly complete scientific pipelines. Lu et al. [2024] show that LLMs can generate ideas, write code, run experiments, and even simulate review processes. That work is valuable because it frames scientific discovery as a closed loop rather than a single generation step. However, the evaluation problem in Lu et al. [2024] is still centered on paper-level review rather than question-level assessment. The system proposed here is narrower and more targeted: it focuses on the quality of the *research question itself*, before the project expands into full experimentation.

The proposal also draws on work in grounded generation and test-time adaptation. Retrieval-augmented generation [Lewis et al., 2020] established the value of combining parametric language models with explicit non-parametric memory. ReAct [Yao et al., 2023] further demonstrated that reasoning and tool use can be interleaved, allowing the model to retrieve targeted evidence when needed instead of relying on a static prompt. These ideas matter because a question assessor should not judge novelty or feasibility purely from memory; it should be able to request additional literature evidence on demand.

Finally, several strands of work motivate inference policies that are richer than single-pass prompting. Self-consistency replaces greedy chain-of-thought decoding with agreement across multiple reasoning paths [Wang et al., 2023]. Self-Refine and Reflexion show that iterative critique and revision can improve quality at test time without weight updates [Madaan et al., 2023, Shinn et al., 2023].

Meanwhile, Kadavath et al. [2022] show that language models can produce useful confidence signals under the right framing, which motivates selective abstention and uncertainty-aware escalation. Taken together, these papers suggest that the right comparison for research-question assessment is not a single static inference rule, but a family of adaptive inference policies with different cost–accuracy trade-offs.

3 Methodology

3.1 Problem formulation

Given a topic description t , the system should output a final research question q^* together with a quality assessment. Rather than modeling quality as a single binary variable, we define a multidimensional score vector

$$\mathbf{s}(q) = (s_{\text{nov}}, s_{\text{sig}}, s_{\text{spec}}, s_{\text{ground}}, s_{\text{feas}}, s_{\text{fals}}),$$

where the dimensions correspond to novelty, significance, specificity, literature grounding, feasibility, and falsifiability. The binary verdict *good* or *bad* is then a projection of this richer representation. The final question is selected by

$$q^* = \arg \max_{q \in \mathcal{Q}} \sum_d w_d s_d(q) - \lambda u(q),$$

where $\mathcal{Q}$ is the set of candidate questions, $u(q)$ is an uncertainty penalty, and the weights w_d can be tuned to domain priorities. This formulation matters because a research question can be excellent along some axes and weak along others, and the system should expose that structure rather than hiding it behind a hard label.

3.2 Stage I: literature-grounded candidate generation

The generation stage begins with **deep-literature-search**, which retrieves and synthesizes literature for topic t . Its output is a report R_t and, when available, a graph-like representation G_t that captures entities, mechanisms, and relations across the retrieved literature. We then use **question-formulator-beta** not to generate a single question, but to produce a small candidate set $\mathcal{Q} = \{q_1, \dots, q_K\}$ under diversity-promoting prompting. Concretely, candidate prompts are conditioned on different desiderata: one prompt emphasizes frontier gaps, another emphasizes bridge hypotheses that connect previously separate literatures, and another emphasizes measurement- or experiment-driven framing.

Each candidate is then normalized into a structured *QuestionSpec*. The purpose of *QuestionSpec* is to force the generated question into the form we ultimately want to evaluate. A *QuestionSpec* records the scientific gap, the proposed mechanism or causal link, the main variables and parameter ranges, the observables or endpoints, the intended mode of evaluation, the conditions for falsification, and a short novelty claim relative to retrieved prior work. This representation is especially important in physics-adjacent domains, where good questions often depend on parameter regimes, symmetries, observables, and exclusion conditions. A structured representation makes the assessor less sensitive to writing style and more sensitive to scientific content.

3.3 Stage II: review-supervised assessment corpus

The assessment module requires supervision that is close to question quality rather than paper quality. The corpus construction therefore proceeds in four steps.

First, we collect OpenReview submissions from public venues that expose PDFs, reviews, and final decisions. Because core physics coverage on OpenReview is sparse, the initial corpus should focus on physics-adjacent venues: scientific machine learning, AI for science, machine learning for physical sciences, computational physics workshops, and relevant tracks at major ML conferences. This choice is deliberate. The goal is not to pretend that OpenReview perfectly represents physics review culture; it is to obtain a large enough pool of public rationales while treating cross-domain transfer to physics as an explicit part of the research problem.

Second, for each paper we extract an *idea-only* representation. The extractor reads the title, abstract, and selected introduction or contribution paragraphs while masking result-heavy content, tables, conclusion language, and score improvements. The output is then converted into the same QuestionSpec schema used for generation. This step is essential because the target of supervision is the quality of the underlying idea or research question, not whether the paper happened to execute it well.

Third, we mine rationales from reviews, meta-reviews, and decision notes. Reviewer text is decomposed into atomic arguments with polarity and dimension tags. For example, a sentence may be mapped to “novelty-positive,” “feasibility-negative,” or “significance-positive.” These atomic arguments seed the initial review bank. Meta-reviews and decision notes receive higher weights because they usually encode a more consensus-like view than any single review.

Fourth, we create targets at several granularities. The coarsest target is accept versus reject. A second target is normalized score within venue and year when ratings are available. A third target is pairwise ranking among papers from the same venue and track. A fourth target is reviewer disagreement, which is used as an uncertainty indicator. In addition, we synthesize *counterfactual hard negatives* by perturbing strong QuestionSpecs: broadening the scope, deleting observables, removing falsification conditions, or collapsing novelty by aligning the question too closely with known prior work. These negatives teach the model to distinguish genuinely weak questions from merely rejected papers.

3.4 Stage III: dynamic ArgumentBankGen for question assessment

The assessor extends ArgumentBankGen in three ways. The first extension is representational. Every argument in the bank is stored not as free text alone, but as a tuple containing its rubric dimension, polarity, provenance, scope, and reliability weight. This makes the bank more interpretable and enables dimension-specific inference.

The second extension is architectural. Instead of a single bank, the assessor maintains three interacting banks:

$$\mathcal{A}(q) = \text{TopK}_{\text{div}}\left(\mathcal{A}_{\text{review}} \cup \mathcal{A}_{\text{lit}}(q) \cup \mathcal{A}_{\text{dyn}}(q)\right).$$

Here $\mathcal{A}_{\text{review}}$ is learned from historical review rationales, $\mathcal{A}_{\text{lit}}(q)$ is derived from the current **deep-literature-search** retrieval and graph for the topic, and $\mathcal{A}_{\text{dyn}}(q)$ is synthesized at inference time from retrieved nearest neighbors, recent literature passages, and contradiction or saturation signals in the local literature graph. The diversity operator prevents the merged bank from collapsing into near-duplicate arguments.

The third extension is in the update rule. In standard HypoGeniC, hard cases trigger generation of new hypotheses from a buffered subset of examples [Zhou et al., 2024]. In the proposed system, hard cases trigger generation of new *arguments* from both the buffered examples and their associated literature summaries, following the spirit of Liu et al. [2025]. This means that when the assessor

Algorithm 1 ARGUMENTBANKGEN

Input: dataset $\mathcal{D}$, seed count m , keep κ , buffer cap $B_{\max}$, horizon T

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1:  $\triangleright$  Bootstrapping: create an initial pool of candidate models
2:  $\mathcal{A} \leftarrow \text{get\_arguments}(\{\mathcal{D}_i : i \leq m\})$ 
3:  $\mathcal{B} \leftarrow \emptyset$   $\triangleright$  Hard-case buffer
4: for each  $(u_\tau, v_\tau) \in \mathcal{D}$  do
5:    $\triangleright$  Focus evaluation on the current best-performing candidates
6:    $\mathcal{A}_{\text{elite}} \leftarrow \{p \in \mathcal{A} : p \text{ is among the top-}\kappa \text{ by score}\}$ 
7:   for each  $p \in \mathcal{A}_{\text{elite}}$  do
8:      $\tilde{v}_\tau^p \leftarrow \text{predict}(p, \tau)$   $\triangleright$  Run inference for this candidate
9:      $\text{update\_score}(p, v_\tau, \tilde{v}_\tau^p)$   $\triangleright$  Online reward/score update
10:  end for
11:   $\triangleright$  If many candidates fail here, treat  $(u_\tau, v_\tau)$  as informative
12:  if  $|\{\text{err}(\tilde{v}_\tau^p) : p \in \mathcal{A}\}| \geq \theta$  then
13:     $\mathcal{B} \leftarrow \mathcal{B} \cup \{(u_\tau, v_\tau)\}$   $\triangleright$  Accumulate difficult examples
14:  end if
15:   $\triangleright$  Periodically expand the pool using buffered hard cases
16:  if  $|\mathcal{B}| = B_{\max}$  then
17:     $\mathcal{Q} \leftarrow \text{get\_arguments}(\mathcal{B})$   $\triangleright$  Propose new candidates from buffer
18:     $\mathcal{B} \leftarrow \emptyset$   $\triangleright$  Reset buffer after mining
19:     $\mathcal{A} \leftarrow \{p : p \in (\mathcal{A} \cup \mathcal{Q}) \text{ is among the top-}\kappa \text{ by score}\}$ 
20:  end if
21: end for
22: return  $\mathcal{A}$   $\triangleright$  Final candidate set
```

repeatedly fails on a topic, it does not merely overfit to more labeled examples; it also asks what the literature says about why those cases are difficult. The reward signal for each argument combines classification accuracy, pairwise ranking quality, calibration, and rationale agreement with human review text.

3.5 Stage IV: adaptive inference beyond a static hypothesis bank

Single-step adaptive inference is a useful baseline, but it should not be the only inference policy. The proposed system therefore uses a staged inference cascade.

At the first stage, the system runs a fast single-step assessment over a compact argument subset. If confidence is high, it returns a verdict immediately. This preserves the low-cost benefits of HypoGeniC-style inference.

At the second stage, low-confidence cases are passed to a two-step policy. The first step identifies which rubric dimensions are uncertain or contested; the second step evaluates the question using arguments selected specifically for those dimensions. This is particularly important for novelty and feasibility, which often require different evidence than specificity or falsifiability.

At the third stage, the system applies self-consistent inference over multiple sampled argument subsets [Wang et al., 2023]. The point is not to collect arbitrary chain-of-thoughts, but to test whether the same verdict is recovered under different evidence slices. If agreement is strong, the verdict becomes more trustworthy.

At the fourth stage, the system performs a targeted literature refresh using ReAct-style retrieval [Yao et al., 2023]. For example, if novelty is uncertain, the model issues a focused retrieval query for closest prior art; if feasibility is uncertain, it retrieves methods and datasets relevant to the claimed

Table 1: **LLM-as-a-judge** results on the held-out test set using GPT-5 mini. This measures how accurately it was able to predict whether a tweet was retweeted or not. The same can be used to better predict if a research question would be accepted or not.

Method	Accuracy (%)
Random	19
Zero-shot w/ rubrics	41
Few-shot w/ rubrics	59
ARGUMENTBANKGEN(no update)	45
ARGUMENTBANKGEN(with update)	68

test. The refreshed evidence is then used to rebuild $\mathcal{A}_{\text{dyn}}(q)$.

At the final stage, if disagreement remains high, the system abstains from hard classification and instead returns a critique. This critique is fed back into **question-formulator-beta** in a Self-Refine/Reflexion-style loop [Madaan et al., 2023, Shinn et al., 2023], producing a revised candidate question that is rescored. In other words, the assessor is not just a gatekeeper; it is also a teacher.

3.6 Stage V: final ranking and closed-loop revision

The end-to-end system should generate several candidate questions, not one, because quality is often revealed comparatively. We therefore rank candidates using both absolute rubric scores and pairwise comparisons. The pairwise comparisons answer a practical question that researchers often care about: if only one question can be pursued, which one is strongest? Pairwise ranking is also a useful hedge against score calibration problems.

Once ranking is complete, the top candidate is revised once or twice using assessor critiques. Each critique is constrained to be actionable: it must either sharpen the mechanism, narrow the scope, identify a missing observable, improve grounding against prior work, or explain why the question appears infeasible or redundant. The final output is thus a package consisting of the revised question, its quality profile, the top supporting arguments, the top opposing arguments, and the literature evidence that grounded the verdict.

3.7 Evaluation plan

The proposal should be evaluated using a time-based split to measure temporal generalization and a cross-venue split to measure domain robustness. Core automatic metrics include binary accuracy, ROC-AUC, pairwise ranking accuracy, expected calibration error, and critique usefulness as measured by post-revision score gain. More importantly, a human evaluation with domain experts should compare the final questions produced by (i) one-shot **question-formulator-beta**, (ii) **question-formulator-beta** plus static single-step assessment, and (iii) the full LAGRANGE loop. The primary human outcomes should be perceived novelty, testability, and willingness to pursue the question as a real project. Ablations should isolate the effect of the literature bank, the dynamic bank, the review bank, and the adaptive inference cascade.

We have run preliminary tests using our own [Evaluator](#)¹ that was tested on how accurately retweets can be predicted. We see in table 1, ARGUMENTBANKGEN with update is preferable to few-shot with rubrics because it does not rely on a fixed prompt alone. Instead, it updates its argument

¹<https://github.com/FirstPrinciplesFoundation/creative-generation>

set as it encounters difficult cases, which leads to a stronger decision rule. **The gain from 59% to 68% suggests that adaptive refinement** captures subtle quality signals better than static few-shot examples, making it more suitable for evaluating research questions.

4 Key Issues the Method Tackles

Separating question quality from paper quality. Paper outcomes are a noisy proxy. A paper may be rejected because of missing experiments or poor exposition even if its core question is strong, and accepted work may contain only incremental questions executed well. By extracting idea-only QuestionSpecs and mining review rationales at the sentence level, the method explicitly tries to supervise *question assessment* rather than paper assessment.

Handling frontier-dependent novelty. Novelty is not a static property. A question that looked original last year may now be redundant. Static hypothesis banks struggle here because they encode generic criteria but not the current state of the literature. The proposed use of **deep-literature-search** and query-conditioned argument generation directly addresses this issue by making novelty and feasibility judgments depend on live retrieval.

Adapting to physics-style questions. Physics questions are often parameterized, symbolic, and constrained by observability or exclusion regions. This can be difficult for generic natural-language classifiers. The QuestionSpec abstraction is designed to address exactly this problem by representing variables, regimes, observables, and falsification criteria explicitly rather than leaving them buried in prose.

Providing actionable, interpretable judgments. A binary label is not enough. Researchers need to know *why* a question is weak and how to improve it. Because the assessor reasons through explicit argument banks with provenance, it can return critiques that are inspectable and revision-ready. This is important for trust as well as utility.

Balancing accuracy with inference cost. Dynamic retrieval and revision are expensive if used for every example. The staged inference cascade addresses this by reserving the most expensive operations for cases where cheaper policies are uncertain. The result is a system that can operate cheaply on easy cases without giving up robustness on hard ones.

5 Why the Method Should Work

The proposal rests on four scientific hypotheses.

H1: reviewer rationales encode durable scientific virtues. Even though review outcomes are noisy, reviewer arguments repeatedly refer to stable criteria such as novelty, significance, clarity of the gap, soundness of the evaluation route, and feasibility. We hypothesize that these criteria transfer from paper review to question assessment once execution-specific content is filtered out. In other words, reviewers are imperfect judges of papers, but often useful teachers of what a promising question looks like.

H2: literature-grounded evidence resolves the local part of quality. Historical review rationales can teach generic criteria, but they cannot know whether a topic has become saturated or newly feasible. We hypothesize that the local literature context supplied by `deep-literature-search` provides exactly the evidence needed to evaluate novelty and feasibility at the frontier. This is the main reason to integrate Liu et al. [2025]-style literature refinement into ArgumentBankGen rather than treating the question assessor as a pure classifier.

H3: dynamic inference will outperform static single-step inference. A static bank is attractive because it is simple, but it is brittle when questions are ambiguous or literature-sensitive. We hypothesize that the staged inference cascade will improve pairwise ranking accuracy and calibration over single-step adaptive inference, especially on temporally held-out topics. The mechanism is straightforward: disagreement triggers either deeper reasoning or targeted retrieval, which should reduce both overconfidence and stale judgments.

H4: closed-loop revision will improve the final question more than filtering alone. Filtering weak questions is useful, but revision is more valuable. We hypothesize that critique-conditioned revision, implemented through `question-formulator-beta` and guided by assessor rationales, will produce stronger final questions than selecting the best initial question alone. This expectation is supported by prior work on self-consistent reasoning and iterative refinement [Wang et al., 2023, Madaan et al., 2023, Shinn et al., 2023].

If these hypotheses hold, the system should outperform one-shot generation and static-bank assessment on both automatic metrics and expert preference. More importantly, it should produce outputs that are useful in practice: questions that are not just well written, but actually worth pursuing.

6 Conclusion

This proposal argues for a shift from one-shot research-question generation to a closed-loop, literature-adaptive generator–assessor–reviser. The generation component builds on existing internal infrastructure, namely `deep-literature-search` and `question-formulator-beta`, while the assessment component extends HypoGeniC-style argument-bank learning with review supervision, current literature grounding, and dynamic inference. The core conceptual claim is simple: a good research question is judged partly by enduring scientific virtues and partly by the current state of the literature, so the assessor must be both historically informed and retrieval aware. By unifying these two forms of evidence, LAGRANGE aims to produce research questions that are more novel, more testable, better grounded, and more actionable than those produced by current generation-only systems. As a research program, it is also attractive because it can be evaluated rigorously: through ablations on bank composition, inference policy, and revision loops, and through expert studies that measure whether the final questions are genuinely worth investigating. For an end-to-end system that must generate and assess research questions in a scientific setting, this is the right level of ambition and the right decomposition of the problem.

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